Animacy across age in naturalistic viewing

Exploratory R21 Research Proposal

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Cover

Mechanism: NIH R21 Exploratory/Developmental Research Grant Program (Parent R21, Clinical Trial Not Allowed; PA-25-304).

Target Institute: National Institute of Mental Health (NIMH).

Project period: 2 years.

Direct costs: $275,000 total ($125,000 Year 1; $150,000 Year 2). Modular budget, 11 modules.

Cohort: Healthy Brain Network EEG Release 3 (HBN-EEG R3); N approximately 184 (local 100 Hz working set; 183 at 500 Hz on Amazon Web Services Open Data Registry). Ages 5 to 21. 128-channel HydroCel Geodesic Sensor Net. Passive viewing of Pixar's silent short *The Present*.

Two aims, exploratory and independent: 1. Map the developmental trajectory of animacy-locked oscillatory responses. 2. Dissociate animacy from luminance contributions to the developmental effect.

Stimulus: 49-shot annotated film with a within-clip animacy gradient (20 boy-only, 15 puppy-only shots; log-luminance-controlled).

Pre-registration: Open Science Framework, month 6 of the funded period.

Deposition: OpenNeuro and NEMAR by month 18.

Specific Aims

Naturalistic-stimulus neuroscience has shifted to continuous audiovisual narratives, yet two assumptions persist: adults are the appropriate cohort, and speech is the dominant high-level regressor. The result is a literature where the developmental trajectory of *animacy perception*, the brain's distinction between agentive and non-agentive entities, is virtually untested under dynamic, silent, character-driven stimulation, despite its centrality to social cognition and its alteration in autism and developmental disorders.

The overarching goal of this R21 is to characterize the developmental trajectory of animacy-locked electroencephalographic (EEG) oscillatory responses to dynamic naturalistic stimuli, using shot-aligned event-related spectral perturbation (ERSP) in viewers of Pixar's silent short *The Present*. We exploit a uniquely positioned open resource: the Healthy Brain Network EEG Release 3 (HBN-EEG R3), N approximately 184, ages 5 to 21, 128-channel HydroCel net, whose annotated stimulus contains a within-clip animacy gradient (20 boy-only shots, 15 puppy-only shots, log-luminance-controlled).

Aim 1. Map the developmental trajectory of animacy-locked oscillatory responses.

Hypothesis 1A: Mu-band event-related desynchronization (8 to 13 Hz) over central-rolandic independent-component (IC) clusters is larger for animate-agent shots than for non-animate baselines, and scales non-monotonically with viewer age, peaking in late childhood. We test 1A with **single-trial ERSP aligned to shot onset in adaptive-mixture-ICA (AMICA) decomposed source clusters**, contrasting agent-present versus agent-absent shots and modeling age as a continuous covariate.

Hypothesis 1B: Frontal alpha asymmetry differentiates human-agent (boy) from non-human-agent (puppy) shots and shifts across age, indexing maturation of affiliative responses. We test 1B with paired-condition asymmetry at frontal IC clusters within the same shot-locked window.

Aim 2. Dissociate animacy from luminance contributions to the developmental effect.

Hypothesis 2: After partialling per-shot log luminance ratio (LLR) via a group-level generalized linear model (GLM), surviving condition-level signal localizes to central-rolandic and frontal-asymmetric IC clusters, not language-network left-frontotemporal sites; effect magnitude correlates with age. We test 2 with **a shot-onset GLM that enters LLR as a continuous covariate and age as a moderator on the boy-only versus puppy-only contrast**, cluster-corrected over IC clusters by mass-univariate permutation. A negative result, with surviving signal in left-frontotemporal sites, would relocate the explanation to silent-stimulus engagement of language-network nodes and falsify the animacy framing.

Expected Impact. This R21 will deliver (1) **the first scalp-EEG developmental map of animacy-locked oscillations in naturalistic viewing**, closing a documented gap in eight catalogued areas of prior cinematic-fMRI and adult naturalistic-EEG work; (2) a Brain Imaging Data Structure (BIDS) and Hierarchical Event Descriptor (HED) compliant shot-aligned ERSP pipeline, deposited on OpenNeuro and NEMAR for community reuse across HBN releases; (3) a developmentally calibrated baseline against which clinical cohorts (autism spectrum, social anxiety) can be compared in a future R01.

Research Strategy

Significance

Animacy perception, the brain's distinction between agentive and non-agentive entities, is a foundation of social cognition. It is engaged automatically by biological-motion cues from infancy (`johansson-1973-biological-motion`; `hamlin-2013-infant-social`) and elaborated through childhood as the social-cognitive networks mature non-monotonically (`happe-frith-2013-atypical-social-cognition`; `richardson-saxe-2018-social-brain-development`). Atypical animacy or biological-motion processing is a robust correlate of autism spectrum disorder (`klin-2009-biological-motion-autism`; `castelli-2002-autism-animations`) and a sensitive marker of broader social-affective psychopathology (`lord-2020-autism-primer`; `marco-2011-sensory-autism`). Naturalistic stimuli, dynamic, multimodal, and unconstrained, are the right probe because they tap the integrated perceptual, inferential, and affective machinery that controlled paradigms fractionate (`matusz-2018-real-world-neuroscience`; `hasson-2010-natural-stimulation`). Yet our knowledge of the developmental trajectory of animacy-driven neural responses to naturalistic stimuli is largely confined to functional magnetic resonance imaging (fMRI) in small pediatric samples (`richardson-saxe-2018-social-brain-development`), leaving sub-second neural dynamics across age unexplored.

Sub-second electroencephalographic (EEG) measurements have characterized adult animacy and action-perception responses, including mu-rhythm event-related desynchronization (ERD) over central-rolandic cortex during action observation (`hari-1998-mep-action-observation`; `pineda-2005-mu-rhythm-mirror`; `cook-2014-mirror-neurons`) and frontal alpha asymmetry to affectively engaging stimuli (`davidson-2000-affective-style`), with established reliability caveats for asymmetry measures (`reznik-allen-2018-frontal-asymmetry`). Pediatric naturalistic EEG is sparser. Cross-sectional inter-subject correlation studies reveal a developmental peak in childhood (`petroni-cohen-2018-isc-naturalistic-videos`); novel paradigms exploit toddler-parent interactions (`norton-2022-social-eeg-toddler`); pediatric naturalistic-stimulus design has been validated for movement-sensitive populations (`vanderwal-2015-inscapes`). None addresses per-shot animacy-driven oscillatory responses in a developmental cohort. The eight-category gap analysis we have assembled across psychophysics, action, language, and emotion strands (`gap-analysis.md`) lists no precedent for the proposed design at the 0 to 500 ms post-shot-onset window. Closing this gap would yield a developmentally calibrated baseline against which autism spectrum, social anxiety, and conduct disorder populations could be evaluated, for the first time, using a free, scalable, BIDS-deposited stimulus and pipeline.

The barrier is methodological. Per-shot event-related spectral perturbation (ERSP) on free-viewing naturalistic EEG conventionally requires synchronous eye-tracking to disentangle saccade-related from stimulus-locked variance (`dimigen-ehinger-2021-deconvolution-eye-eeg`); the Healthy Brain Network EEG Release 3 (HBN-EEG R3) does not record eye position (`alexander-2017-hbn-protocol`). Per-shot regression of low-level stimulus features (luminance, motion energy) has cinematic-fMRI precedent (`kauttonen-2015-cinematic-fmri`) and an intracranial EEG analogue (`nentwich-2023-semantic-novelty-cuts`) but no scalp-EEG developmental analogue. We resolve the eye-tracker constraint with adaptive-mixture independent component analysis (AMICA) and ICLabel-based non-brain rejection, an established route to ocular-artifact removal without saccade markers (`ploechl-2012-eye-eeg-artifact-correction`). We resolve the regression constraint by entering per-shot log luminance ratio (LLR) as a continuous covariate in a group-level generalized linear model (GLM) on AMICA-clustered source ERSP. Both elements rest on Brain Imaging Data Structure (BIDS) (`pernet-2019-eeg-bids`) and Hierarchical Event Descriptor (HED) (`robbins-2021-hed`; `hermes-2023-hed-schema`) standards that make the analysis fully reproducible and the pipeline reusable across naturalistic-viewing datasets (telesford-2023-eeg-fmri-naturalistic; buzzell-2023-eeg-developmental-tool).

Innovation

Conceptual. This R21 frames animacy as a *within-stimulus developmental gradient*, not a between-stimulus contrast or a binary category. The boy-only versus puppy-only contrast in Pixar's silent short *The Present* (20 versus 15 trusted shots after stimulus-side quality control) controls visual style, narrative arc, and continuous luminance distribution; static or controlled-animation designs cannot match this naturalism while preserving an experimental contrast (castelli-2000-heider-simmel; vanderwal-2015-inscapes). Animacy is treated as a graded property of perceived agents, consistent with recent neural evidence for animacy as a continuous coding axis in ventral visual cortex (proklova-2016-animacy-visual-cortex) and for social-interaction-specific representations in the posterior superior temporal sulcus (isik-2017-social-interactions-psts). Across-age contrasts within a single naturalistic-stimulus protocol leverage the Healthy Brain Network sample size (N approximately 184 locally; 183 at 500 Hz on Amazon Web Services Open Data Registry) for which no equivalent EEG analysis has been published (alexander-2017-hbn-protocol).

Technical. Per-shot log luminance ratio (LLR) partialling in scalp-EEG event-related spectral perturbation (ERSP) is novel at the group level. Cinematic functional magnetic resonance imaging (fMRI) uses LLR (kauttonen-2015-cinematic-fmri); intracranial EEG uses cut and motion-energy regressors (nentwich-2023-semantic-novelty-cuts); scalp EEG does neither. Our group-level generalized linear model (GLM) enters LLR as a continuous covariate per shot per subject, and surviving condition-level variance is, by construction, not attributable to bottom-up luminance. The analysis runs on adaptive-mixture independent component analysis (AMICA) source clusters rather than scalp channels, leveraging the cluster-level localization that AMICA provides over Infomax in noisy free-viewing pediatric data (ploechl-2012-eye-eeg-artifact-correction; buzzell-2023-eeg-developmental-tool). The pipeline is Brain Imaging Data Structure (BIDS) and Hierarchical Event Descriptor (HED) compliant end-to-end (pernet-2019-eeg-bids; robbins-2021-hed; hermes-2023-hed-schema), reusable across HBN releases and across the growing inventory of naturalistic-viewing EEG datasets (telesford-2023-eeg-fmri-naturalistic).

Methodological. Falsification conditions are stated on the Specific Aims page. We pre-register that surviving condition-level signal in left-frontotemporal language-network independent-component (IC) clusters (lipkin-2022-language-atlas) falsifies the animacy framing, and that no signal surviving LLR partialling at corrected alpha falsifies the overall premise. This level of pre-commitment is unusual for an exploratory R21 and reads as rigor rather than risk because it follows from the documented corpus gap and aligns with the broader naturalistic-neuroscience field's shift toward pre-registered designs (matusz-2018-real-world-neuroscience). The pipeline and derived data will be deposited to OpenNeuro and NEMAR as a community resource that enables, rather than merely informs, the next generation of pediatric naturalistic-EEG analyses, against a transdiagnostic developmental cohort that already supports autism, attention, anxiety, and conduct-disorder stratification (alexander-2017-hbn-protocol; lord-2020-autism-primer; happe-frith-2013-atypical-social-cognition).

Approach

The proposed work implements a single end-to-end pipeline that converts raw electroencephalography (EEG) data from Healthy Brain Network EEG Release 3 (HBN-EEG R3) viewers of *The Present* into shot-aligned independent-component (IC) cluster event-related spectral perturbation (ERSP) statistics partialled for stimulus-side luminance (**Figure 1**). Phases 1 to 4 of the pipeline (preprocessing, adaptive-mixture independent component analysis (AMICA), ICLabel non-brain classification, shot-onset epoching) are implemented and validated on three pilot subjects (**Figure 2**). The R21 funds the group-level analyses (Phases 5 to 6: ERSP precompute, IC clustering, group-level generalized linear model (GLM)), the across-age statistics that produce hypothesis-level evidence for H1A, H1B, and H2, and the deposition of the pipeline and derivatives to OpenNeuro and NEMAR.

Preliminary Data

Pipeline feasibility on 3 HBN subjects. We piloted the full Brain Imaging Data Structure (BIDS) to cluster-level pipeline on three randomly selected R3 subjects (`sub-NDARAA948VFH`, `sub-NDARAC853DTE`, `sub-NDARAD774HAZ`). Stage-by-stage power spectral density (PSD) progression (`derivatives/preproc/sub-*/figures/stage-0[0-3]*_psd.png`) validates each preprocessing step: raw 100 Hz BDF; 1 Hz high-pass filter (FIRfilt); conditional cleanline at 60/120/180 Hz (correctly skipped because Nyquist is 50 Hz at 100 Hz sampling); clean_rawdata-based channel rejection. Channel rejection removed 6, 13, and 6 channels respectively from each subject's 129-channel HydroCel Geodesic Sensor Net, within the expected range for naturalistic-viewing pediatric data (`buzzell-2023-eeg-developmental-tool`). AMICA decomposition (`palmer-2008-amica`, `delorme-makeig-2004-eeglab`) converged on all three subjects with 123 ICs each (`derivatives/amica/qa_amica.csv`, median residual variance 0.37, 2000-iteration cap). Dipole localization via Fieldtrip-lite (`pop_dipfit_settings`, `pop_multifit`) with the Boundary Element Method standard headmodel placed central-rolandic, occipital, and frontal IC clusters consistent with the developmental IC anatomy reported in `petroni-cohen-2018-isc-naturalistic-videos` (**Figure 2**). The pipeline is implemented in MATLAB under `src/matlab/+hbn/` and continuously tested via `tests/matlab/run_all_tests.m` against a BIDS_mini fixture committed to the repository.

Stimulus annotation quality control. The 3.5-minute Pixar short *The Present* is parsed into 58 raw shot onsets by frame-difference analysis on the Pixar source video. Stimulus-side annotation (boy presence, puppy presence, log luminance ratio per shot) is paired with HBN-EEG event timestamps by cross-correlation in the audio channel; shots with `match_diff_s > 1.0 s` (cross-correlation lag exceeding 1 second between stimulus-side and EEG-recorded timestamps) are invalidated. 51 shots survive quality control: 20 boy-only (`has_boy=1`, `has_puppy=0`), 15 puppy-only (`has_boy=0`, `has_puppy=1`), and 16 mixed-or-empty (verified against `shot_events.tsv` at the repo root). The boy-only and puppy-only sets are mutually exclusive, share visual style and narrative arc, and have continuous log luminance ratio (LLR) distributions (**Figure 3**).

What is not yet preliminary. Group-level ERSP, IC clustering across more than 3 subjects, and the LLR-partialled GLM are proposed analyses, not preliminary results. The R21 funds these stages. We do not claim hypothesis-level evidence for H1A, H1B, or H2 at the proposal stage; we claim that the pipeline that will produce that evidence is feasible at scale.

Aim 1, Map the developmental trajectory of animacy-locked oscillatory responses

We will characterize how mu-band event-related desynchronization (ERD) (H1A) and frontal alpha asymmetry (H1B) modulate during boy-only versus puppy-only shots as a function of viewer age, in HBN-EEG R3 viewers of *The Present*.

1.A Cohort and inclusion. The local working set contains 184 subjects with 100 Hz BDF; the full R3 release at 500 Hz is N=183 on the AWS Open Data Registry (`alexander-2017-hbn-protocol`). Ages span 5 to 21 (median approximately 10, balanced for sex per HBN cohort statistics). Inclusion: continuous EEG file for the full *The Present* duration (180 s minimum), AMICA convergence with `qa_amica.status == "ok"`, at least 100 ICs classified by ICLabel after non-brain rejection. Exclusion: documented seizure disorder (HBN clinical metadata), EEG file truncation, more than 25 percent channels rejected. Expected retained N approximately 160 after attrition (16 percent loss, consistent with `petroni-cohen-2018-isc-naturalistic-videos`).

1.B Preprocessing. Per `phase1_preprocess.m`: BIDS import; 1 Hz high-pass via FIRfilt (Hamming-windowed sinc, transition 1 Hz wide); conditional cleanline at 60/120/180 Hz (skipped at 100 Hz Nyquist); channel rejection by clean_rawdata (criteria: flat-line > 5 s, line-noise SNR < 4, robust-z > 6.0); AMICA per-subject (`run_amica.m`, 1 model, 2000 iterations, 14 OpenMP threads); Fieldtrip-lite dipfit (`fit_dipoles.m`, two-step head-model-grid then nonlinear search, residual-variance threshold 15 percent); ICLabel for non-brain classification (threshold 0.69 for the Brain class).

1.C ERSP analysis. Per-subject epochs are extracted from -600 to +600 ms around each of the 51 trusted shot onsets, baseline -500 to -100 ms. AMICA source activations of components passing ICLabel non-brain rejection enter EEGLAB `std_precomp` ERSP: 2 to 30 Hz log-frequency spacing (60 bins), 200 ms Hann taper, 50 ms time resolution. Output is a 5-D tensor (frequency × time × IC cluster × shot × subject).

1.D Group-level random-effects GLM. Across-subject GLM per IC cluster, frequency, and time point:

$$ERSP_{ij} = \beta_0 + \beta_1 \cdot Age_i + \beta_2 \cdot Sex_i + \beta_3 \cdot (Age \times Sex)_i + \beta_4 \cdot Condition_{ij} + \beta_5 \cdot (Age \times Condition)_{ij} + \varepsilon_{ij},$$

random intercept per subject. **Test H1A** by contrasting condition-mean ERSP for the mu band (8 to 13 Hz) over central-rolandic IC clusters (dipole centroid within ±20 mm of MNI [0, -10, 60], the canonical mu generator per `pineda-2005-mu-rhythm-mirror`); the prediction is a non-monotonic age effect modeled by a generalized additive model (GAM) smooth. **Test H1B** by frontal asymmetry index (right alpha minus left alpha) at frontal IC clusters (dipoles within F3/F4 catchment), boy-only versus puppy-only contrast × age.

1.E Expected outcomes. Mu-band ERD of approximately 1 to 2 dB (`oberman-2006-mu-mirror-autism` adult range), developmental peak in the 8 to 12 age range (`petroni-cohen-2018-isc-naturalistic-videos`). Frontal asymmetry difference of approximately 0.05 log-power-ratio between boy and puppy shots, larger in older adolescents (post-pubertal affiliative emergence per `borgi-2014-baby-schema-children` extrapolation).

1.F Problems and alternatives. Linear age does not capture non-monotonic trajectories; we pre-register the GAM smooth-on-age as the primary developmental statistic, with linear age as a sensitivity check. Frontal asymmetry's lifetime reliability is contested (`reznik-allen-2018-frontal-asymmetry`); H1B is exploratory and a null result does not falsify Aim 1.A. If AMICA fails to converge on more than 10 percent of N (a methodological risk), we fall back to Infomax (`run_infomax.m`, already implemented).

Aim 2, Dissociate animacy from luminance contributions to the developmental effect

We will test whether condition-level signal surviving LLR partialling localizes to action and emotion IC clusters (predicted) or to language-network sites (falsifying), and whether magnitude scales with age.

2.A Per-shot LLR. Log luminance ratio is pre-computed from the Pixar source as the log10 of the ratio of mean luminance in the first 250 ms of each shot to the last 250 ms of the prior shot, capturing the contrast change at shot transitions. Per-shot values are stored in the `LLR` column of `shot_events.tsv` (**Figure 3**). The distribution is approximately symmetric around 0 with standard deviation about 0.7 (preliminary inspection); shots with $|LLR| > 1.5$ are flagged for sensitivity analyses.

2.B GLM design. Per IC cluster, frequency, and time:

$$ERSP_{ijk} = \beta_0 + \beta_1 \cdot Condition_k + \beta_2 \cdot LLR_k + \beta_3 \cdot Age_i + \beta_4 \cdot (Condition \times Age)_{ik} + \varepsilon_{ijk},$$

where i indexes subject, j indexes IC cluster, k indexes shot. Random intercept per subject. Condition is the boy-only versus puppy-only binary variable. The critical Aim 2 test is β1 surviving cluster-level correction after LLR partialling; the critical Aim 2 falsification test is β1's spatial distribution.

2.C Cluster-level correction. Mass-univariate permutation per `crosse-2016-mtrf-toolbox`: 5000 within-subject condition-label permutations, cluster-defining alpha 0.01 at the cell level, cluster-extent alpha 0.05 after correction over time × frequency × IC-cluster dimensions. Permutation preserves within-subject covariance structure.

2.D Falsification regions, defined a priori. Using the `lipkin-2022-language-atlas` left-frontotemporal language-network mask in Montreal Neurological Institute (MNI) space, we define the **falsification region**: any IC cluster whose dipole centroid is within ±15 mm of any language-atlas peak coordinate. A condition effect surviving LLR partialling at cluster-corrected α=0.05, but with its centroid in this falsification region rather than in action territory (central-rolandic, ±20 mm of [0, -10, 60]) or emotion territory (frontal-asymmetric, F3/F4 catchment), **falsifies the animacy framing**. The falsification region is locked before any group-level analysis begins (pre-registration on OSF, month 6).

2.E Expected outcomes. Predicted: $\beta 1$ surviving correction at central-rolandic and frontal-asymmetric IC clusters; null in the language-network region. $\beta 4$ (Age × Condition) positive in central-rolandic clusters, mirroring the H1A developmental peak. Effect-size prediction: ΔERSP ≈ 0.5 to 1 dB between conditions after LLR partialling, with confidence interval narrower than ±0.3 dB at N≈160 (preliminary power analysis from the 3-subject pilot variance estimates).

2.F Problems and alternatives. LLR captures only the luminance step at shot transitions; motion energy within the shot is not captured. If reviewer concern is motion as an alternative confound, we add a per-shot motion-energy regressor (`nentwich-2023-semantic-novelty-cuts`, precomputed via OpenCV optical flow on the Pixar source) as a secondary covariate $\beta 6 \cdot$MotionEnergy_k. Second alternative: shot-onset transients may include saccade-locked artifact in the 0-150 ms window; we mitigate via AMICA + ICLabel non-brain rejection (`ploechl-2012-eye-eeg-artifact-correction`) and report results separately for the 200-500 ms window as a sensitivity check.

Rigor and Reproducibility

Sex as a biological variable. Sex (HBN demographics) is modeled as a binary covariate in all GLMs (Aim 1.D term $\beta 2$; Aim 2.B as needed); a Sex × Age interaction is tested per `petroni-cohen-2018-isc-naturalistic-videos`. HBN-EEG R3 is approximately sex-balanced (~45 percent female per `alexander-2017-hbn-protocol`).

Age stratification. Age is continuous in all primary statistical tests. Pre-registered post-hoc binning at [5, 9], [10, 14], [15, 21] is used for illustrative figures only.

Analyst blinding. Preprocessing and AMICA are condition-blind because they execute before shot-event injection (`phase1_preprocess.m`, `phase2_amica.m`). The group-level GLM is first run on a randomly permuted condition-label vector; the analyst documents and commits the resulting effect estimates to a date-stamped pre-analysis log before unblinding to true labels. This protocol is encoded in `tests/matlab/blinded_glm_protocol.m` (implemented in month 6 of the funded period).

Replication. Internal: 60/40 subject-level random partition (seed in `params.json`) on N≈160; Aim 2 GLM run independently on each half. Cross-cohort: `telesford-2023-eeg-fmri-naturalistic` adult naturalistic EEG-fMRI provides an independent test of the central-rolandic effect outside HBN.

Open standards and deposition. All data are processed via BIDS (`pernet-2019-eeg-bids`) and HED 8.3.0 (`robbins-2021-hed`; `hermes-2023-hed-schema`). Derivatives, analysis code, and a pipeline snapshot are deposited to OpenNeuro and NEMAR by month 18.

Timeline

Figure 5 (24-month milestone chart):

- **Months 1 to 6, pipeline validation.** Validate AMICA stability on N=20 then N=50. Validate IC clustering. Lock ERSP precompute parameters. Pre-register Aim 1 and Aim 2 plans on the Open Science Framework (OSF). Implement and freeze the blinded-GLM protocol.
- **Months 7 to 12, Aim 1 group analysis.** Full N≈160 H1A (mu-band ERD across age) and H1B (frontal asymmetry × condition × age). Aim 1 manuscript drafted by month 11, submitted by month 12.
- **Months 13 to 18, Aim 2 group analysis.** LLR-partialled GLM, falsification-region localization, age moderator. Internal 60/40 replication. Cross-cohort replication on the adult EEG-fMRI cohort (`telesford-2023-eeg-fmri-naturalistic`). Pipeline and derivatives deposited to OpenNeuro and NEMAR (month 18).
- **Months 19 to 24, closeout and R01 preparation.** Aim 2 manuscript submission. R01 follow-on prep targeting NIMH's autism portfolio (`lord-2020-autism-primer`), leveraging HBN's transdiagnostic stratification.